2014-2015 – Master 2 – Macro I

Lecture 8 : the Mortensen-Pissarides matching model

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1. A simple framework

The matching function

- The basic building block is the matching function, which relates hirings per unit of time to the two key inputs in the search process, unemployment and vacancies :

$$H_t = m(U_t, V_t). \tag{1}$$

- Here $H_t =$ the gross hiring rate per unit of time, $U_t =$ the number of unemployed workers, $V_t =$ the number of vacant jobs.
- The matching function is similar to a production function, and we assume it has the same properties including constant returns to scale.
- Note that in this framework, unemployment and vacancies are not a waste : they are a productive input in the production of new matches.
- This defines the process for job creation. To begin with, we assume a simple process of job destruction : a fraction s of all jobs is destroyed per unit of time.

1. A simple framework

Evolution of unemployment

- Let $\bar{L}$ = the total labor force, L_t = employment at t. Then we can define the hiring, unemployment, and vacancy rates in relation to the total workforce :

$$\begin{aligned} u_t &= \frac{U_t}{\bar{L}} = \frac{\bar{L} - L_t}{\bar{L}}, \\ v_t &= \frac{V_t}{\bar{L}}, \\ h_t &= \frac{H_t}{\bar{L}}. \end{aligned}$$

- Because of constant returns to scale, we can rewrite (1) as

$$h_t = m(u_t, v_t).$$

- The evolution of the unemployment rate is

$$\begin{aligned} \frac{du}{dt} &= -h_t + s(1 - u_t) \\ &= -m(u_t, v_t) + s(1 - u_t). \end{aligned}$$

- This defines a $du/dt = 0$ locus in the (u, v) plane which is called the "Beveridge curve".
- Along this locus, we have

$$\begin{aligned} 0 &= -m'_u du - m'_v dv - s du \\ &\Longrightarrow \frac{dv}{du} = -\frac{m'_u + s}{m'_v} < 0. \end{aligned}$$

- Furthermore,

$$\begin{aligned} \frac{d^2 v}{du^2} &= -\frac{\left(m''_{uu} + m''_{uv}\frac{dv}{du}\right) m'_v - (m'_u + s)(m''_{uv} + m''_{vv}\frac{dv}{du})}{m'^2_v} \\ &\propto (-m''_{uu}m'_v) + (m''_{vv}\frac{dv}{du})(m'_u + s) + (m''_{uv}(m'_u + s - \frac{dv}{du}m'_v)), \end{aligned}$$

and all the terms in parentheses in the last expression are > 0, therefore

$$\frac{d^2 v}{du^2} > 0.$$

- Note : "$\propto$" means "proportional with the same sign".

1. A simple framework

Labor market tightness

- Convexity of the Beveridge curve : because decreasing marginal returns to each input in the matching function.
- When I increase vacancies by one unit when vacancies are large, the effect on hirings is small, and only a small reduction in unemployment would maintain a balance between employment outflows and inflows.
- Given constant returns, it is easier to think in terms of labor market tightness rather than vacancies. By definition, labor market tightness is

$$\theta = v/u.$$

- The probability per unit of time of finding a job is

$$p = h/u = \frac{m(u,v)}{u} = m(1,\theta) = p(\theta), p' > 0, p'' < 0.$$

- The probability per unit of time of filling a vacancy is

$$q = \frac{m(u,v)}{v} = m(\frac{1}{\theta},1) = q(\theta), q' < 0.$$

- Furthermore,
$$p(\theta) = \theta m(\frac{1}{\theta}, 1) = \theta q(\theta).$$
- The Beveridge curve can be reexpressed in the (u, θ) plane :
$$\begin{aligned} \dot{u} &= s(1-u) - up(\theta) \\ &= s(1-u) - u\theta q(\theta). \end{aligned}$$
- Along this curve :
$$\begin{aligned} \frac{d\theta}{du} &= -\frac{s+p}{up'} < 0; \\ \frac{d^2\theta}{du^2} &\propto -up'^2 \frac{d\theta}{du} + (s+p)(p' + p'' u \frac{d\theta}{du}) > 0. \end{aligned}$$

- The Beveridge curve delivers one dynamic relationship between u and v (or θ). Above it vacancies are larger than in steady state, so unemployment is falling. Below it, unemployment is rising. Hence the arrows on Figure 1.
- To complete the model we need another relationship between u and θ. This will come from labor demand.

1. A simple framework

The Beveridge curve

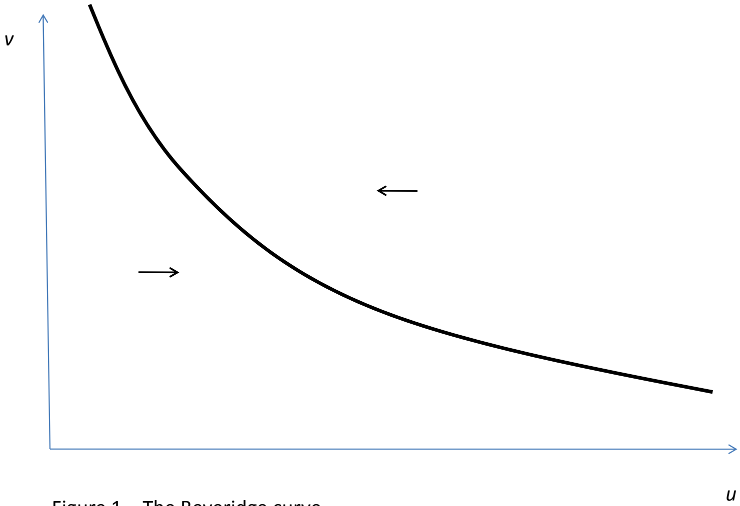

Figure 1 – The Beveridge curve

- We assume there is a single homogeneous good. Once a worker finds a job, he produces a constant flow of this good equal to y per unit of time. He is paid a fixed wage w. There is a fixed real interest rate equal to r. Let J_t be the value of the firm at t. Since the job is destroyed with flow probability s, the asset valuation equation for J is

$$rJ = y - w + \dot{J} - sJ. \tag{2}$$

- The only non explosive solution is

$$J = \frac{y - w}{r + s}.$$

- To recruit workers, firms must post vacancies. Posting a vacancy costs c per unit of time. Let V_v be the value of a vacancy. Its asset valuation equation is

$$rV_v = -c + q(\theta)(J - V_v) + \dot{V}_v.$$

- There is free entry in posting vacancies. Therefore,

$$V_v = 0.$$

- Thus we get

$$J = \frac{c}{q(\theta)}.$$

- Note that the expected duration of a vacancy is $1/q(\theta)$, therefore this tells us that the value of a job is equal to the average recruiting cost per job.
- This determines the equilibrium value of θ, which is constant and equal to

$$\theta = q^{-1}\left[\frac{c(r+s)}{y-w}\right].$$

- Figure 2 shows the adjustment dynamics.

1. A simple framework

Equilibrium

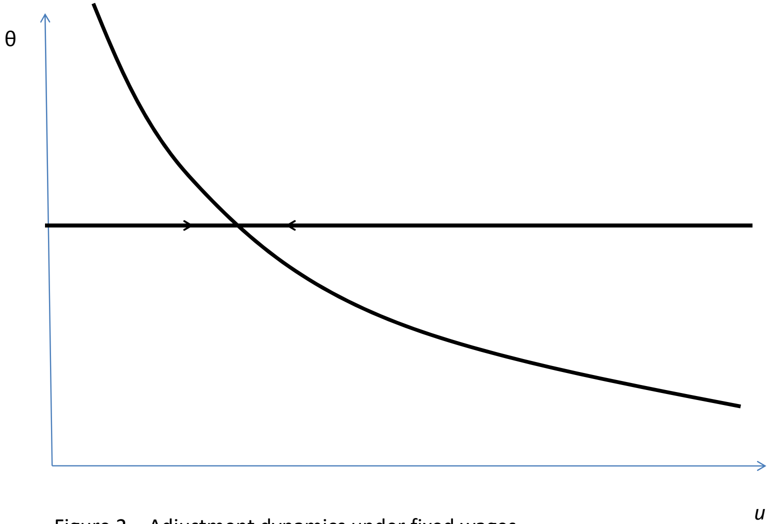

Figure 2 – Adjustment dynamics under fixed wages

- θ goes up, and u falls, if the profitability of a job goes up, i.e. if r goes down, y goes up,w goes down.
- θ goes up if the cost of a vacancy falls.
- All these changes do not affect the BC. Thus the economy moves along the BC. (Figure 3)
- A rise in s shifts both the labor demand curve and the BC through the discounting and mechanical effects of job destruction. (Figure 4)
- Assume shocks to y alternate : this suggests that business cycles induce counter-clockwise loops around the Beveridge curve.

1. A simple framework

Comparative statics

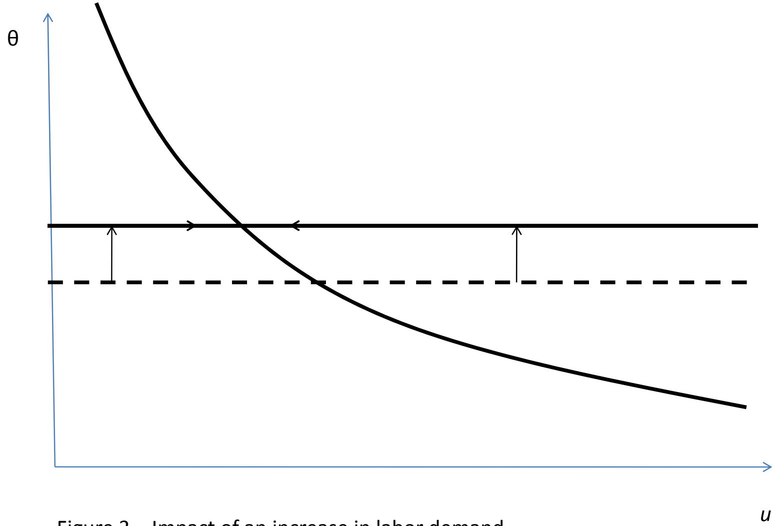

Figure 3 – Impact of an increase in labor demand

1. A simple framework

Comparative statics

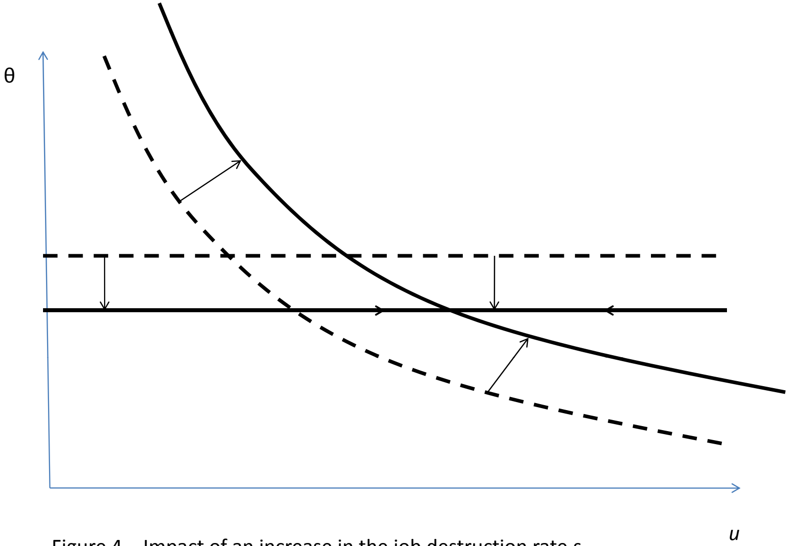

Figure 4 – Impact of an increase in the job destruction rate *s*

- An elegant way to endogenize wages is to assume that incumbent employees and employers bargain over the surplus created by the sunk recruiting costs.
- Bargaining is individual between each worker and the firm. It is easiest to assume that 1 firm = 1 job.
- Let V_e be the value of being employed, V_u be the value of being unemployed.
- The asset valuation equations for V_e and V_u are (assuming no unemployment benefit)

$$\begin{aligned} rV_e &= w + s(V_u - V_e) + \dot{V}_e; \qquad (3) \\ rV_u &= \theta q(\theta)(V_e - V_u) + \dot{V}_u. \qquad (4) \end{aligned}$$

- It is convenient to use the net surplus of the match, i.e.

$$W = J + V_e - V_u.$$

- Consolidating (2),(3), and (4) we get

$$rW = y - \theta q(\theta)\varphi W - sW + \dot{W}. \tag{5}$$

- At each date wages are set so as to maximize the joint log Nash product :

$$\ln\left[(J - V_v)^{1-\varphi}(V_e - V_u)^{\varphi}\right].$$

- Bargaining takes place at the firm-worker level, taking as given aggregate conditions V_u and V_v.
- This is because the bargaining sets the wage *within* the match, not for all the economy
- Recall that $J = \frac{y-w}{r+s}$
- Recall that $rV_e = w + s(V_u - V_e) + \dot{V}_e$ so that the non-explosive solution is $V_e = \frac{w+sV_u}{r+s}$

- For any increase in wages Δw we have (all else equal) $\Delta V_e = \frac{1}{r+s}\Delta w$ and $\Delta J = -\frac{1}{r+s}\Delta w = -\Delta V_e$.
- Therefore the FOC is :

$$\frac{1-\varphi}{J-V_v} = \frac{\varphi}{V_e - V_u}.$$

- Since $V_v = 0$, this is equivalent to

$$V_e = V_u + \frac{\varphi}{1-\varphi} J. \tag{6}$$

- That is :

Value of being employed = Outside option + rent.

- Since $J = \frac{c}{q(\theta)}$, the rent is proportional to the total recruiting cost that has been spent.
- Equation (6) determines the wage despite that the wage does not explicitly appear in it.
- We now need to derive a relationship between θ and u in this more complicated model.

- Note that (6) implies that this net surplus is shared in proportion $(\varphi, 1-\varphi)$, that is

$$
\begin{aligned}
V_e - V_u &= \varphi W, \\
J - V_v &= J = (1-\varphi) W.
\end{aligned}
$$

$$rW = y - \theta q(\theta)\varphi W - sW + \dot{W}. \tag{5}$$

- The term in $\theta q(\theta)\varphi W$ is the opportunity cost to the worker of being employed in this match instead of being looking for another job which would yield a net value φW to the worker and have an arrival rate $\theta q(\theta)$.
- Last, W can be expressed as a function of θ, since

$$\frac{c}{q(\theta)} = J = W(1-\varphi).$$

- We get a dynamic equation for θ :

$$\frac{(r+s)c}{(1-\varphi)q(\theta)} = y - \frac{c}{(1-\varphi)q(\theta)^2}q'(\theta)\dot{\theta} - \frac{\theta\varphi c}{1-\varphi}. \tag{7}$$

- The $\dot{\theta} = 0$ schedule is such that θ is constant and solves

$$\frac{(r+s)c}{(1-\varphi)q(\theta)} = y - \frac{\theta\varphi c}{1-\varphi}.$$

- Since $q' < 0$, (7) has unstable dynamics locally around $\dot{\theta} = 0$.
- As θ is a non-predetermined variable, we pick the only non-explosive solution, i.e. the saddle-path which is horizontal (Figure 5).
- The adjustment dynamics are qualitatively the same as in the case where w is fixed.

2. Endogenous wages

Dynamics

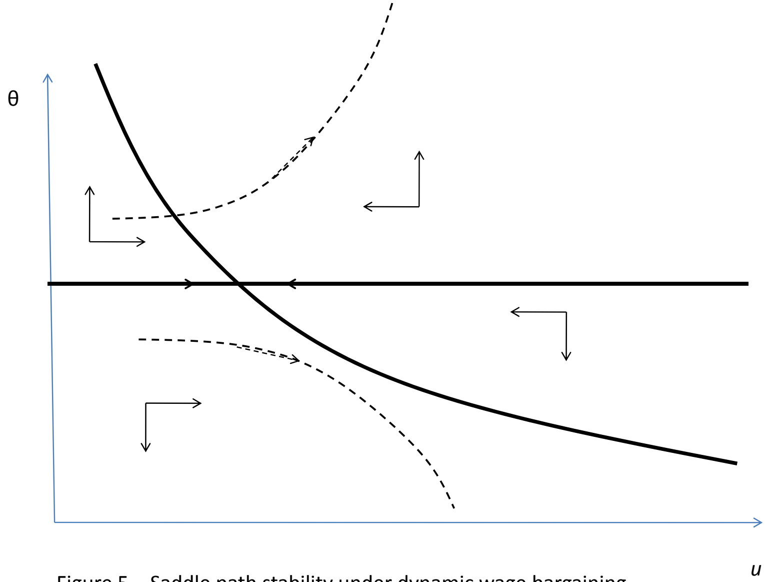

Figure 5 – Saddle path stability under dynamic wage bargaining